Journal of Sol-Gel Science and Technology (2023) 105:337–347
https://doi.org/10.1007/s10971-022-06019-0

ORIGINAL PAPER: DEVICES BASED ON SOL-GEL OR HYBRID MATERIALS

The morphology effect of embedded ZnO particles-based composite on flexible hybrid piezoelectric triboelectric nanogenerators for harvesting biomechanical energy

P. Paydari[1] · N. Manavizadeh[2] · A. Hadi[1] · J. Karamdel[1]

Received: 13 August 2022 / Accepted: 12 December 2022 / Published online: 21 December 2022
© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2022

Abstract

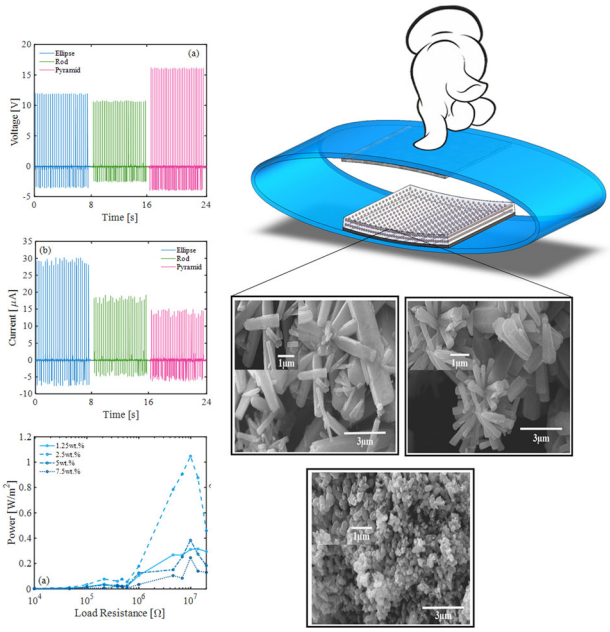
Harvesting mechanical energy with a flexible and weightless device for self-power electronics has attracted many interests, nowadays. Hybrid nanogenerators by combining different power generating techniques enhance the performance of nanogenerators. Zinc oxide as a non-toxic ceramic has significant piezoelectric properties. In this paper, piezo/triboelectric hybrid nanogenerators based on three different shapes of ZnO nanoparticles are fabricated. The impact of the ellipse, rod, and pyramid-shaped ZnO particles encapsulated in PDMS matrix on the hybrid nanogenerators performance is investigated. XRD analysis shows pyramid-shaped ZnO particles have more crystals in the c-axis direction which is the piezoelectric effect direction in ZnO material. SEM results show the average size of particles from biggest to the smallest is rod, pyramid, and ellipse shapes, respectively. The maximum open-circuit voltage and short-circuit current are obtained from the pyramid and ellipse-shaped ZnO particle/PDMS composite hybrid nanogenerators with the value of 161 V and 29.8 μA, respectively. The maximum power density of the ellipse, rod, and pyramid-shaped ZnO particle/ PDMS composite-based hybrid nanogenerators is about 1.04, 0.75, and 0.6 Wm^{-2}, respectively, occurred at 10 MΩ load resistance. In addition, different capacitors are charged by proposed nanogenerators demonstrating the PDMS embedded pyramid-shaped ZnO particles nanogenerator produced greater voltages in the capacitances. Consequently, the hybrid nanogenerators based on the ellipse and pyramid-shaped ZnO particle/ PDMS composites have better performances due to the structure and morphology properties of piezoelectric ZnO particles and can be utilized for the power source of milliwatt flexible portable devices.

✉ N. Manavizadeh
manavizadeh@kntu.ac.ir

[1] Faculty of Electrical Engineering, Islamic Azad University South Tehran branch, Tehran, Iran

[2] Faculty of Electrical Engineering, K.N. Toosi University of Technology, Tehran, Iran

Graphical Abstract



Keywords Tribo/Piezo hybrid nanogenerators · Flexible nanogenerator · Biomechanical harvester · ZnO particles · PDMS composite · Pyramid-shaped particle

Highlights

- Harvesting mechanical energy device for self-power electronics has attracted many interests.
- The impact of three different shapes of ZnO particles encapsulated in the PDMS matrix on the performance of piezo/triboelectric hybrid nanogenerators was investigated.
- Results demonstrated the maximum current and voltage values occurred in 2.5 wt.% wight ratio of ZnO particles in the PDMS matrix.
- The maximum voltages obtained from HNGs based on the ellipse, rod, and pyramid-shaped ZnO particles were about 11.95, 10.6 and 16.1V.
- The maximum power density under load resistance generated by the ellipse, rod, and pyramid-shaped ZnO particle composite-based HNGs was about 1.04, 0.75, and 0.6 Wm-2, respectively at 10 MΩ.

1 Introduction

Technological and economical progression is one of the major impacts of the industrial revolution. The globalization process has been increasing the demand for electricity as the number of industries power source [1–3]. Regarding dwindling finite fossil fuel resources as the central energy resources, at a rapid rate, harvesting energy from renewable energy sources such as solar, wind, and hydropower is a valuable approach to provide global energy needs [4]. Exploiting energy harvesting devices not only is a solution for the energy crisis issue but also plays a significant role in poverty alleviation and human survival [5, 6]. On the way to modern society, daily life in many aspects is based on the interaction between humans and smart devices. Thanks to the internet of things (IoT) technologies, smart devices can communicate with each other in addition to humans to facilitate our daily life [7–9]. Owing to the widespread usage of personal portable devices and sensors, attention to sustainable self-power devices for personalizing biomedical health monitoring, human motion sensing and humanoid robotics emerge. Hence, a self-power system that harvests the energy from human movements and vibrations appeals to researches for the real-time and long-term functioning of monitoring devices [10, 11]. Biomechanical energy scavenging from the human body is an energy studies spotlight due to its sustainability and environmentally friendly nature. A desired wearable energy harvester is a flexible and weightless device capturing a wide range of movement such as finger tapping, breathing, and body movement to power electronic devices or to be used for sensor networks [12–16].

Efficiently harvesting different types of mechanical energy which is the most abundant and accessible energy in our daily activities is demonstrated by nanogenerators [17]. Amongst all types of nanogenerators, triboelectric (TENG) and piezoelectric (PENG) nanogenerators present significant potential due to their unique benefits [18, 19]. Triboelectric nanogenerators work based on triboelectrification and electrostatic induction principles [19, 20]. Through the coupling of these principles mechanical energies of the ambient environment, for instance; biomechanical energy, wind energy, and water wave energy can be effectively and endlessly converted to electricity by TENGs [21–25]. Advantages such as broad material availability, small volume, low cost, low-frequency response, and relatively high electrical output enable TENG ideally suited for powering IoT and other self-power sensors using biomechanical energy [26–29]. On the other hand, piezoelectric nanogenerators are introduced as one of the promising approaches for harvesting ambient mechanical vibration, stress, or strain by taking the merits of the electromechanical coupling effect in inherent piezoelectric materials [19, 30]. Although piezoelectric nanogenerator is appropriate for a large force with small deformation, its lightweight, low cost, and easy manufacturing make it a suit nominate for vibration-electricity conversion. Piezoelectricity can be seen in many different categories of materials, such as ceramics, single crystals, and polymers. Piezoelectric ceramics including ZnO, $BaTiO_3$ (BTO), and PZT have attracted interest in piezo generators [19, 31–34].

Zinc Oxide (ZnO) is a wurtzite crystal type, non-ferroelectric material which shows polarization along the c-axis [35, 36]. Although Zinc Oxide has a low piezo-electric constant rather than that for PZT and BTO, its low dielectric (ε_r) and moderate piezoelectric strain constant (e) leads to a higher piezoelectric voltage constant [37–39]. Moreover, ZnO can be synthesized and fabricated in a diversity of nanostructures which makes it feasible for flexible piezo-energy harvesting devices and sensors [40, 41]. The vertical ZnO nanowires, routinely grow along the c-axis in the hydrothermal synthesis method which is the same as polarization direction, as a result, it improves the output performance of ZnO-based PENGs [35, 42]. It is worthy to mention that in addition to high piezoelectricity, due to the low cost, abundance, tunability, ease of fabrication, and nontoxicity, ZnO stands out for wearable and implantable energy harvesting devices [43, 44]. While other piezoceramics are toxic because of Pb content, Different forms of Zinc oxide including nanoarrays and nanoparticles particularly is noted as an excellent biocompatible material [45–47].

Materials biocompatibility is one of the key factors in wearable electronics. Polydimethylsiloxane (PDMS) is introduced as an ideal biocompatible material for flexible wearable electrical sensors and energy harvesters. Inherent elasticity makes PDMS discretionarily twisted, rolled, folded, compressed, and stretched [16]. While ZnO has fragile and non-deformability features, a combination of ZnO with a flexible organic material such as PDMS has drawn increasing attention in flexible biocompatible devices [18]. Embedding inorganic filler into a polymer is a special approach to prepare a new composite category for enhancing functionality with wide applications. This strategy merges the unique electrical, magnetic, optical, thermal, and piezoelectric properties of inorganic particles at the micro/nanoscale with the features of the polymer matrix, including processibility, flexibility, and stability [48].

Tribo-piezoelectric hybrid nanogenerators (HNG) can overcome the challenge of high electrical output performance of nanogenerators [49]. Although piezoelectric nanogenerators have low output voltage, the piezo-harvesting technique can be applied to various places with less influence by an external condition such as humidity and temperature [18, 50]. On the other aspect, the stable output voltage performances of triboelectric nanogenerators with no degradation can be preserved under various deformations, whereas its low output current has been a great challenge for exploiting it widely [51, 52]. Hybrid nanogenerators can be a great strategy for high-power flexible wearable devices. Bringing triboelectricity and piezoelectricity effect together by utilizing a composite film enhances the overall output performance due to the electron transfer improvement and induction of additional charge across the active material which is termed as hybrid nanogenerator [53]. Herein, three hybrid piezo/triboelectric nanogenerators based on ZnO nanoparticles were designed and fabricated. The main notion of this study was investigating the impact of different shapes of ZnO nanoparticles on hybrid nanogenerator performance. Spherical, pyramid, and rod-shaped nanoparticles which were synthesized hydrothermally were considered in this probe. Field-effect scanning electron microscopy (FESEM) and X-ray diffraction analysis were employed to study the structural and morphological properties of nanoparticles. The effect of ZnO nanoparticle content on PDMS composite film as well as the thickness of composite film on the output voltage and current performance of nanogenerators were investigated. Furthermore, the output power of the nanogenerators was studied in different load resistance. Furthermore, the sharp ZnO nanoparticle/PDMS composite film presented better performance owing to the cooperation of triboelectric and piezoelectric effects.

2 Experimental detail

ZnO/PDMS composite was fabricated with Different shapes of ZnO nanoparticles which were synthesized by the

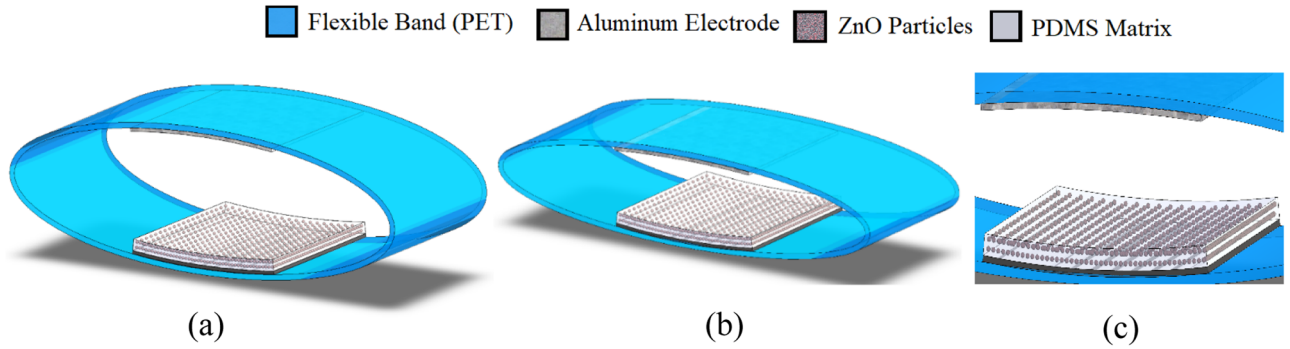


Fig. 1 schematic structure of ZnO particle/PDMS composite based hybrid nanogenerator in two cases **a** release and **b** under force

hydrothermal precipitation method. The high-purity grade materials Zinc nitrate hexahydrate ($Zn(NO_3)_2{\cdot}6H_2O$, Merk), Hexamethylenetetramine (HMTA, Merk), and ammonia solution 25% (Merk) were utilized. For synthesizing of spherical nanoparticle 0.47 Mol of Zinc nitrate was dissolved in DI water and heated up. When the temperature of the solution reached 60 °C, Ammonia solution was added dropwise while the solution was stirring. Then the solution was kept at 60 °C for 30 mins. Finally, the sediment powder was filtered, immersed with Ethanol and DI water several times, and dried at 80 °C for 6 h. In the case of pyramid-shaped particles, equimolar Zinc nitrate and HMTA (0.01 Mol) were individually dissolved in DI water. The growth solution was kept at a constant temperature of 90 °C for 7 h with a magnetic stirrer. Then the solution with its sediments was centrifuged and filtered to collect sediments while immersing several times and finally dried at 80 °C for 6 h. Moreover, rod-shaped nanoparticles were synthesized by hydrothermal method with 11.4 mMol of Zinc nitrate and HMTA which were individually dissolved in the solvent. The growth solution was stirred with magnet while kept at a constant temperature of 90 °C for 2.5 h. Then similar to the aforementioned particles, particles were collected by utilizing filter paper, then immersed, finally dried at 80 °C.

After the synthesis of ZnO particles, the PDMS composition was fabricated. At the outset, the PDMS polymer was fabricated by mixing a silicone elastomer base and a silicone elastomer curing agent with a 10:1 ratio well. Then the synthesized ZnO particles with different shapes were added into the PDMS matrix with different weight ratios in three pairs of samples. After that, the PDMS with ZnO particles were mixed homogeneously following by transferring it into a petri dish, and heated at 100 °C for 1 h in the oven. At the final step, the resultant film was taken out from the dish and trimmed into 2 × 2 cm^2 dimensions to fabricate nanogenerators.

For fabricating tribo/piezoelectric hybrid nanogenerators, the aluminum tape was opted to be an electrode due to its accessibility, cost-effectivity, and fabrication simplicity. While ZnO/PDMS composite film served as a tribo-negative material, the aluminum tape was considered as a tribo-positive material as well as electrode. The composite films were attached to the bottom Al electrodes and finally, square-shaped hybrid nanogenerators were prepared by fastening the two electrodes to face each other exploiting a PET-based strip which is an appropriate material for flexible mechanical nanogenerators (Fig. 1).

To investigate morphological and structural attributes of different shapes of ZnO particles synthesizing by hydrothermal, field emission scanning electron microscopy (FESEM) images were obtained using Hitachi S-4160 (20 kV, Japan) and X-ray diffraction (XRD) analysis was performed by Philips Expert pro using Cu-kα radiation ($\lambda = 1.5418$ Å). Moreover, Electrical characterizations of tribo/piezoelectric HNGs were performed by a digital oscilloscope (DSO 2100, MEGATEK).

3 Results and discussion

Utilizing microparticles-based composites not only permits the facilitation of fabrication on any scale, but also it is a simple premise to enhance the piezoelectric effect. Crystal directions of particles is one of the aspects that influence piezoelectric characteristics. The crystallinity and structure direction of ZnO particles in different synthesized shapes were analyzed by XRD and the results are represented in Fig. 2. The obtained diffraction pattern of ZnO ellipse, rod, and pyramid-shaped particles agreed with the JCPDS (no.79-0208) data and ICDD-PDF database [54, 55]. In all shapes of ZnO particle, the highest peaks reflected from 36.5° which means most of the crystallites grow along (101) direction. The pyramid-shaped ZnO sample has significantly more particles in the (002) direction rather than the other shapes which is an important direction for the piezoelectric effect. In rod-shaped particles, most crystallites grow on (101), (100), and (110), respectively. In the case of ellipse-shaped particles, expect (101) direction, the

abundance of the other crystal directions is seen approximately equivalent. Moreover, the size of particles has a considerable impact on piezoelectricity features [48]. The crystallite size obtained from XRD analysis is the mean coherent domain size. Hence, the morphology study is an important issue for investigating particles implemented by scanning electron microscopy characterization. Figure 3 depicts FESEM analysis of three shapes synthesized ZnO particles. According to SEM images, the average size of ellipse-shaped particles is about 0.28 µm and the smallest and the largest particle size are approximately 0.14 and 0.55 µm. In addition, the rod-shaped particle size in length is varied from 1.13 and 7 µm with an average of 3.53 µm. Although the thickness of rods, in this case, is between 0.26 to 1.9, these particles can be invoked as microparticles. Moreover, Fig. 3c shows the pyramid-shaped ZnO particles which the smallest and the largest length of these particles are 1.24 and 4.48 µm, respectively, with an average length of 2.3 µm while the diameter of pyramid sections of these particles differs between 0.33 and 1.56 µm and can be called microparticles as well.

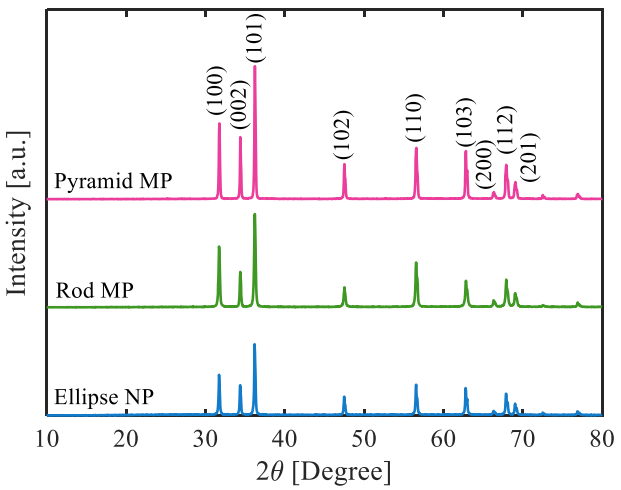


Fig. 2 XRD patterns spectra of Ellipse, Rod and Pyramid- shaped ZnO particles

To assess the synergetic triboelectric and piezoelectric effect on the HNG, experimentally, the different weight ratio of ZnO particles encapsulated in the PDMS solution was studied by measuring V_{OC} and J_{SC} curves under an applied force and frequency of 5 N and 4 Hz, respectively. Figure 4 represents the output open-circuit voltage of different ZnO particle shapes composite hybrid nanogenerators. The electrical output performance of hybrid nanogenerators based on ZnO particles embedded in PDMS is probed in different weight ratios of 1.25, 2.5, 5, and 7.5 wt %. The output voltage of the ellipse-shaped particle-based nanogenerator is depicted in Fig. 4a. By increasing the concentration of particles from 1.25 to 2.5 wt.% the open-circuit voltage enhances to approximately 120 V. Nevertheless, the higher concentration of the particles leads to a decline in the output voltage. As it can be seen from Fig. 4b, c, the decrement in the open-circuit voltage by augmentation in the concentration of rod and pyramid-shaped ZnO particles more than 2.5 wt.% occurs. The maximum output voltage of rod and pyramid-shaped ZnO microparticle composite-based nanogenerators are 106 and 161 V, respectively which was obtained in 2.5 wt.% concentration. Moreover, the V_{OC}s of 2.5wt.% ZnO particles: PDMS composite HNG in the pyramid and ellipse-shaped particles are significantly higher than that in the rod-shaped particle-based composite. The slight difference in the output voltage of the rod-shaped particle composite in different weight ratio concentrations can be a result of the morphological and crystal structure of the rod particle sample which influences the piezoelectric effect. In addition to size deviation in the rod-shaped particle sample that affects piezo-coefficient, the XRD results indicate the dominant crystal directions of most of the particles in rod-shaped were not along the c-axis which is the main crystal direction of wurtzite ZnO producing piezoelectric-charge. Hence the alternation in the amount of rod-shaped particles

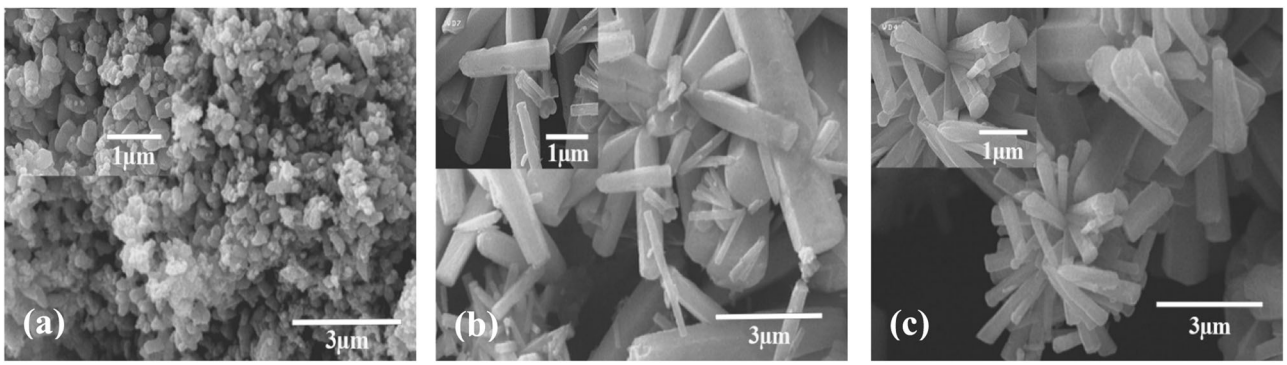


Fig. 3 Morphology study by FESEM analysis of **a** Ellipse-shaped, **b** Rod-shaped, and **c** Pyramid-shaped ZnO particles

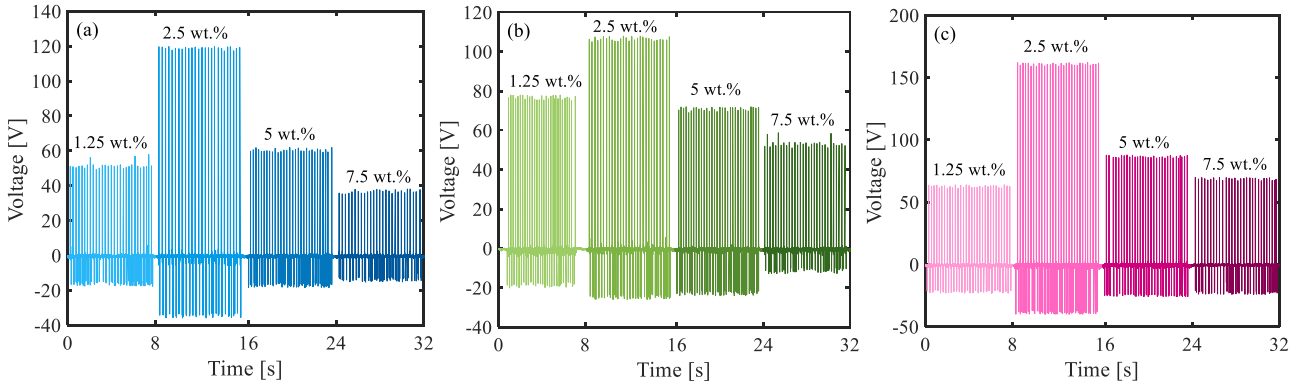


Fig. 4 Open circuit voltage of **a** Ellipse, **b** Rod, and **c** Pyramid-shaped ZnO particle/PDMS composite-based piezo-triboelectric hybrid nanogenerators

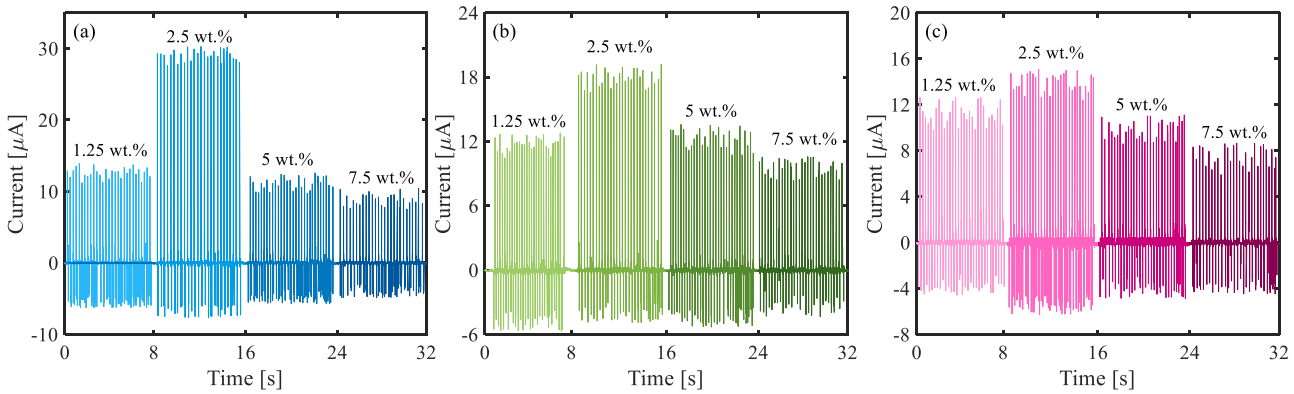


Fig. 5 Short circuit current of hybrid nanogenerators based on **a** Ellipse, **b** Rod, and **c** Pyramid-shaped ZnO particle/PDMS composite

concentration in the composite did not have a major change in charge generation and voltage production as well.

Furthermore, short circuit current is another key parameter of nanogenerators presented in Fig. 5. The output current of the ZnO particle composite-based hybrid nanogenerator raises with the increment in the particle concentration from 1.25 to 2.5 wt.%. However, the short circuit currents decline by the concentration higher than 2.5 wt.%. In contrast to the V_{OC}, the current of pyramid-shaped ZnO particle composite-based HNG does not have much dependence on particle concentration in the composite in comparison with other particle shapes owing to the structural features of pyramid-shaped particles and can be inferred to a few changes in polarization charge density. The pyramid-shaped ZnO particle has a higher density rather than the aforementioned particles, as a result, the composite samples based on pyramid particles are more transparent. Thus, the augmentation in the amount of mass in 5 and 7.5 wt.% cases, does not increase the volume of the particle in PDMS solution significantly and the derivative of generated dipoles electric charge as a consequence current does not alter remarkably.

The augmentation in the electrical output performance of piezo-triboelectric HNGs based on ZnO particle/PDMS composite was noticed due to the enhancement in relative permittivity of the ZnO/ PDMS composite. Moreover, the surface charge density of the composite film was raised by embedding ZnO particles in the PDMS matrix. Nonetheless, further increasing the weight ratio of ZnO particles in PDMS solution from 2.5 to 7.5wt.% make the performance of the HNGs reduce.

The V_{OC} and J_{SC} of ZnO composite-based nanogenerators incremented with extension the weight ratio of ZnO particle content in the PDMS polymer, owing to the Si-O-Si bond-breaking effect on the charge produced by the tribo-effect. Regarding the high sensitivity of PDMS film to the free oxygen radicals under a mechanical force, generally, the charge generating by applying force alternates. The Si-CH_3 groups split, and the reactive oxygen radicals generated by ZnO particles attack some of the Si-CH_3 groups and are substituted with the Si-OH group [56]. Hence, the breaking bonds on the PDMS composite film surface are activated to be charged and the recurring dangling bonds cause the way of generating the electric charges. Consequently, the increase

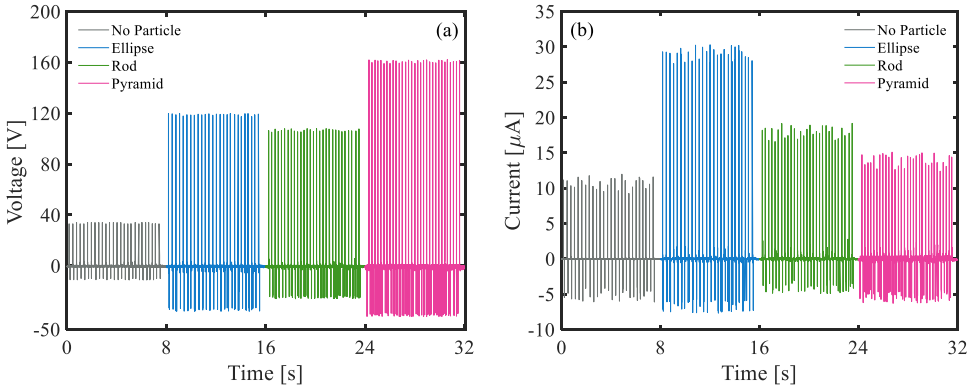


Fig. 6 Comparison of **a** V_{OC}, and **b** I_{SC} of HNGs based on Ellipse, Rod and Pyramid-shaped ZnO particle in PDMS matrix with the weight ratio of 2.5 wt.% in comparison with pure PDMS tribo-nanogenerator

in the ZnO particle content of the PDMS composite makes the surface charge density raise along with the dangling bonds with the result of HNG performance boosting. Nevertheless, when the ZnO particle weight ratio in PDMS matrix is further increased from 2.5 wt.% to 5 and 7.5 wt.%, the bond breaking of Si-O-Si groups gets saturated. Thus the performance of composite-based HNG declined as the ZnO particle concentration in composite increased further [37].

Figure 6 represents the maximum output voltages and currents of hybrid nanogenerators which were obtained in 2.5 wt% of ZnO particle in the PDMS matrix case in comparison with pure PDMS triboelectric nanogenerator. The open voltage of pyramid-shaped ZnO particle-based composite nanogenerator is about 160 V which is higher than the V_{OC} of the ellipse and rod-shaped particle-based HNGs. Moreover, the maximum short-circuit current belongs to the HNG based on the ellipse-shaped particle composite which is about 29.8 μA. The J_{SC} of 2.5wt.% rod and pyramid-shaped ZnO particle/PDMS composite-based nanogenerator 18.9 and 14.6 μA, respectively. The augmentation in voltage and current of hybrid nanogenerator in comparison to pure PDMS TENG with 35 V and 11 μA output determines the beneficial effect of ZnO piezoparticles on the performance of HNGs. According to piezoelectric studies on different ZnO one-dimensional nanostructures, the thinner the edge of the nanostructure is, the more charges generate due to the piezoelectric effect [57]. Thus, the pyramid-shaped ZnO microparticles generate more charges followed by higher voltage in HNG based on pyramid-shaped particles. In addition, regarding the morphological structure of the aforementioned particles, the ellipse-shaped particles were in nm size, while the size of rod and pyramid-shaped particles were in μm. As a result of the high surface energy of ZnO nanoparticles, they easily precipitated out of solution in the high volume which significantly limits improvement in the dielectric and piezoelectric responses [48]. The surface charge density on the ellipse-shaped ZnO particles-based composite, dipole density, and thus the output current of HNG is higher than the other particles-based nanogenerators.

The other critical parameter in composite-based nanogenerators is the thickness of the tribo-insulator. The impact of PDMS-based composite thickness on the nanogenerator performance is explored. Figure 7a illustrates the output voltage of hybrid nanogenerators based on three different shapes of ZnO particles with 2.5wt.% content of ZnO particles in PDMS matrix. The samples for this investigation were fabricated with the double thickness (Thickenss was about 1 mm) of the regular samples. Results indicate the output performance of thicker composite-based nanogenerators is poorer rather than thinner ones. The V_{OC} of the thick nanogenerators follows the trend of thinner nanogenerators, which means the voltage of pyramid-shaped particle composite has the highest voltage of 45 V, and the ellipse and the rod-shaped particle-based composite generated lower voltages as 34 and 23 V, respectively. As depicted in Fig. 7b, the current of thicker composite nanogenerators based on pyramid-shaped particles is the greatest with the amount of approximately 1 μA. This issue can be described by the morphological properties of the particles and the generated tribo-charges on the surface of the composite described later. Moreover, by the increase in the thickness of the tribo-insulator the induced charges on the electrodes decrement, hence, the voltage and the current decreased. This phenomenon was observed in the triboelectric nanogenerator based on pure PDMS as well.

Furthermore, the output power in different load resistance is a momentous parameter to evaluate the nanogenerator practically. The power density of hybrid nanogenerators based on different shapes of ZnO particle composite under different load resistance is illustrated in Fig. 8. The output current of all fabricated hybrid nanogenerators was declined with increasing external load resistance from 10 KΩ to 22 MΩ. The maximum output power density was obtained utilizing Eq. (1);

$$w = \frac{R_L I^2}{A} \tag{1}$$

where R_L expresses load resistance, I indicates the current flow across the resistance, and A is the area of the composite.

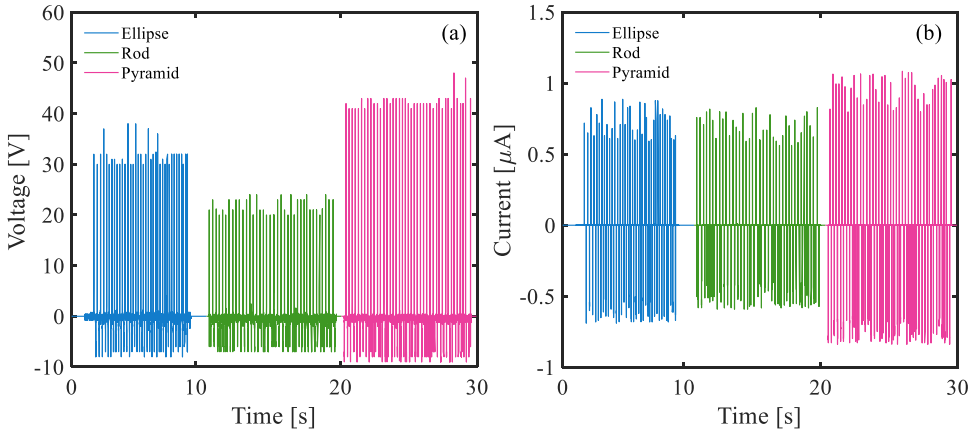


Fig. 7 **a** V_{OC}, and **b** I_{SC} of dual thick PDMS composite film-based HNGs with 2.5 wt.% content of different shapes of ZnO particles

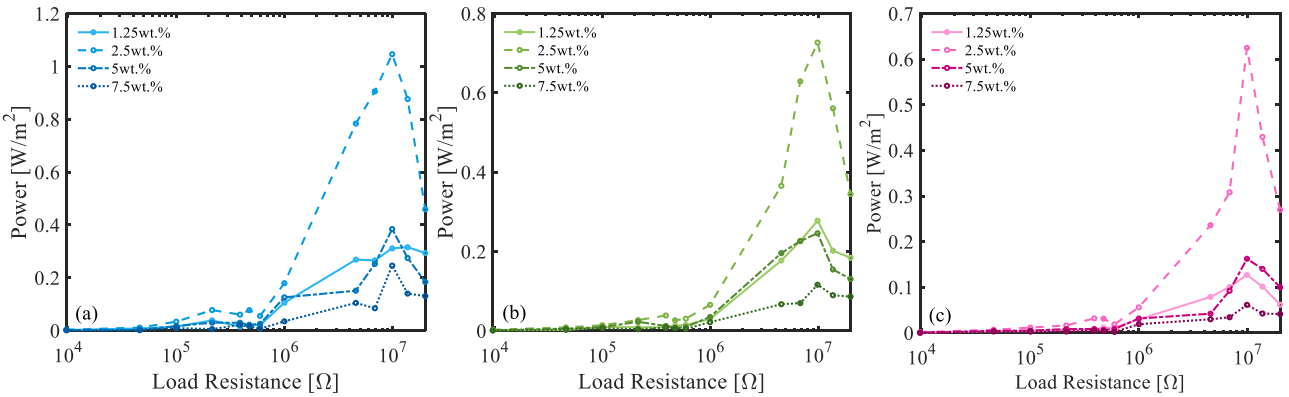


Fig. 8 Power density of different weight ratio of **a** Ellipse, **b** Rod, and **c** Pyramid-shaped ZnO particle/PDMS composite- based HNGs

The maximum output power of all ZnO particle composite-based HNGs was generated at 10 MΩ. Figure 8 represents, generated power of the hybrid nanogenerators based on 2.5 wt.% ZnO particle content in the PDMS matrix is significantly higher for greater load resistance in all different shapes of particles. The generated power of an ellipse-shaped particle-based composite nanogenerator is depicted in Fig. 8a which the maximum produced power is about 1.04 Wm^{-2}. The maximum power generated by rod and pyramid-shaped ZnO particle/ PDMS HNGs is approximately 0.75 and 0.6 Wm^{-2}, respectively (Fig. 8b, c). Results indicate ellipse-shaped particle/PDMS composite-based HNG generates greater power at the same load resistance due to the higher current production. It is worth mentioning that the current generated by ellipse-shaped particle composite in short circuit mode without load resistance is greater than the rod and pyramid-shaped particle composites. Generally, regarding the alternating current (AC) of the electrical output of the hybrid nanogenerators, their output cannot be directly exploited to charge capacitors or batteries. Hence, a full-wave bridge rectifier was utilized to convert the AC output signal of HNGS to the direct current signal (DC). The nanogenerators connected with the full rectifier were then exploited to charge various capacitors including, 0.5, 1, 3.3, 10.7, 20.57, and 42 μF, which illustrated in. Figure 9 illustrates obtained capacitances charging curves, indicating the fabricated HNGs can efficiently charge capacitors. The charging process of capacitors was studied for 60 seconds. In this period, the 0.5 μF capacitor has a greater stored voltage of about 4.6 V in pyramid-shaped particle composite-based HNG owing to the generation of higher voltage (Fig. 9c) that was investigated previously. The obtained curves affirm that the pyramid-shaped ZnO particle/PDMS-based hybrid nanogenerator can be employed as a self-power source for portable electronics. Since proposed hybrid nanogenerators can generate an electrical signal in response to the mechanical forces, self-power sensing applications can exploit these HNGs [58]. The nanogenerator-based sensors have some merits in comparison to the other structure of sensors such as capacitive and resistive due to the independence of sensors to an external power source. The proposed pyramid-shaped ZnO particle /PDMS composite-based HNG demonstrates the ability of harvesting low-frequency biomechanical energy from daily human activities and can be utilized in electrical automobiles or underneath pedestrian ways, as well.

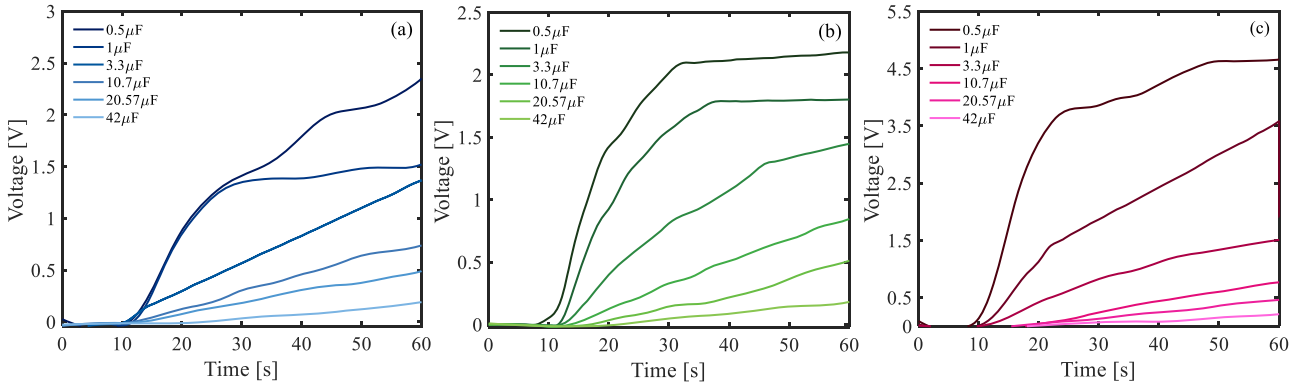


Fig. 9 Charging curves of various capacitors using the full wave rectifier circuit with 2.5 wt.% content of **a** Ellipse, **b** Rod, and **c** Pyramid-shaped ZnO particle in PDMS matrix

4 Conclusion

Embedding piezoelectric material in the insulator matrix is a great approach to enhance the piezoelectric coefficient of the insulator and fabricate piezo/ triboelectric hybrid nanogenerators. In this paper, the impact of three different shapes of ZnO particles encapsulated in the PDMS matrix on the performance of piezo/triboelectric hybrid nanogenerators was investigated. ZnO particles in three shapes including ellipse, rod and pyramid were synthesized by the hydrothermal method. XRD analysis results expressed that pyramid-shaped nanoparticles had more particles in the c-axis direction which is important for piezo-coefficient. Moreover, the size of ellipse-shaped ZnO particles was in the 500 nm range, whereas the average particle size of rod and pyramid-shaped particles was in the range of 3.5 and 2.3 µm, respectively. The open-circuit voltages and short-circuit currents of three cases fabricated HNGs were probed in different weight ratio content of ZnO particles in PDMS solution. Results demonstrated the maximum current and voltage values occurred in 2.5 wt.% wight ratio of ZnO particles in the PDMS matrix. The maximum voltages obtained from HNGs based on the ellipse, rod, and pyramid-shaped ZnO particles were about 120, 106 and 160 V. Although the I_{SC} of ellipse-shaped particle composite was the greatest at about 29.8 µA, this current was about 18.9 and 14.6 µA for rod and pyramid-shaped particle composite-based HNGs, respectively. Furthermore, the thinner PDMS and composite films had greater generated voltage and current in both triboelectric nanogenerators and hybrid nanogenerators owing to the relative dielectric coefficient. For more analysis, the maximum power density under load resistance generated by the ellipse, rod, and pyramid-shaped ZnO particle composite-based HNGs was about 1.04, 0.75, and 0.6 Wm^{-2}, respectively at 10 MΩ. In addition, different capacitances were charged by the full-wave rectifier connected to the proposed HNGs. The capacitances reached higher voltages in the charging process by pyramid-shaped particle composite-based hybrid nanogenerators due to the nanogenerator's produced voltage. The hybrid nanogenerators based on the ellipse and pyramid-shaped ZnO particles composites can be exploited for wearable and portable devices owing to the generated power, current, and voltage by regarding the application.

Compliance with ethical standards

Conflict of interest The authors declare no competing interests.

References

1. Kaltschmitt M, Streicher W, Wiese A (2007) Renewable energy: technology, economics and environment. Springer Science & Business Media
2. Panwar NL, Kaushik SC, Kothari S (2011) Role of renewable energy sources in environmental protection: A review. Renew Sustain energy Rev 15(3):1513–1524
3. Sen S, Ganguly S (2017) Opportunities, barriers and issues with renewable energy development–A discussion. Renew Sustain Energy Rev 69:1170–1181
4. Philibert C (2017) Renewable energy for industry. International Energy Agency, Paris
5. Mahapatra B, Patel KK, Patel PK (2021) A review on recent advancement in materials for piezoelectric/triboelectric nanogenerators. Mater Today: Proc 46:5523–5529
6. Zhang Y, Gao X, Wu Y, Gui J, Guo S, Zheng H, Wang ZL (2021) Self-powered technology based on nanogenerators for biomedical applications. Wiley Online Library, pp 90–114
7. Qiu C, Wu F, Shi Q, Lee C, Yuce MR (2019) Sensors and control interface methods based on triboelectric nanogenerator in IoT applications. IEEE Access 7:92745–92757
8. Liu L, Shi Q, Ho JS, Lee C (2019) Study of thin film blue energy harvester based on triboelectric nanogenerator and seashore IoT applications. Nano Energy 66:104167
9. He Q, Li X, Zhang J, Zhang H, Briscoe J (2021) P–N junction-based ZnO wearable textile nanogenerator for biomechanical energy harvesting. Nano Energy 85:105938

10. Pu X, Liu M, Chen X, Sun J, Du C, Zhang Y, Zhai J, Hu W, Wang ZL (2017) Ultrastretchable, transparent triboelectric nanogenerator as electronic skin for biomechanical energy harvesting and tactile sensing. Sci Adv 3(5):e1700015
11. Xie M, Hisano K, Zhu M, Toyoshi T, Pan M, Okada S, Tsutsumi O, Kawamura S, Bowen C (2019) Flexible multifunctional sensors for wearable and robotic applications. Adv Mater Technol 4(3):1800626
12. Shi Q, He T, Lee C (2019) More than energy harvesting–Combining triboelectric nanogenerator and flexible electronics technology for enabling novel micro-/nano-systems. Nano Energy 57:851–871
13. Stadlober B, Zirkl M, Irimia-Vladu M (2019) Route towards sustainable smart sensors: ferroelectric polyvinylidene fluoride-based materials and their integration in flexible electronics. Chem Soc Rev 48(6):1787–1825
14. Choi Y-M, Lee MG, Jeon Y (2017) Wearable biomechanical energy harvesting technologies. Energies 10(10):1483
15. Sezer N, Koç M (2021) A comprehensive review on the state-of-the-art of piezoelectric energy harvesting. Nano Energy 80:105567
16. Kim DH, Dudem B, Yu JS (2018) High-performance flexible piezoelectric-assisted triboelectric hybrid nanogenerator via polydimethylsiloxane-encapsulated nanoflower-like ZnO composite films for scavenging energy from daily human activities. ACS Sustain Chem Eng 6(7):8525–8535
17. Zhang X, Si Y, Mo J, Guo Z (2017) Robust micro-nanoscale flowerlike ZnO/epoxy resin superhydrophobic coating with rapid healing ability. Chem Eng J 313:1152–1159
18. Zhao Z, Dai Y, Dou SX, Liang J (2021) Flexible nanogenerators for wearable electronic applications based on piezoelectric materials. Mater Today Energy 20:100690
19. Singh HH, Kumar D, Khare N (2021) A synchronous piezoelectric–triboelectric–electromagnetic hybrid generator for harvesting vibration energy. Sustain Energy Fuels 5(1):212–218
20. Yang Y, Zhang H, Chen J, Jing Q, Zhou YS, Wen X, Wang ZL (2013) Single-electrode-based sliding triboelectric nanogenerator for self-powered displacement vector sensor system. Acs Nano 7(8):7342–7351
21. Xi Y, Guo H, Zi Y, Li X, Wang J, Deng J, Li S, Hu C, Cao X, Wang ZL (2017) Multifunctional TENG for blue energy scavenging and self-powered wind-speed sensor. Adv Energy Mater 7(12):1602397
22. Feng Y, Zhang L, Zheng Y, Wang D, Zhou F, Liu W (2019) Leaves based triboelectric nanogenerator (TENG) and TENG tree for wind energy harvesting. Nano Energy 55:260–268
23. Wang Y, Yang E, Chen T, Wang J, Hu Z, Mi J, Pan X, Xu M (2020) A novel humidity resisting and wind direction adapting flag-type triboelectric nanogenerator for wind energy harvesting and speed sensing. Nano Energy 78:105279
24. Wu H, Wang Z, Zi Y (2021) Multi-mode water-tube-based triboelectric nanogenerator designed for low-frequency energy harvesting with ultrahigh volumetric charge density. Adv Energy Mater 11(16):2100038
25. Wang J, Jiang Z, Sun W, Xu X, Han Q, Chu F (2022) Yoyo-ball inspired triboelectric nanogenerators for harvesting biomechanical energy. Appl Energy 308:118322
26. Yun Y, La M, Cho S, Jang S, Choi JH, Ra Y, Kam D, Park SJ, Choi D (2021) High quality electret based triboelectric nanogenerator for boosted and reliable electrical output performance. Int J Precis Eng Manuf-Green Technol 8(1):125–137
27. Cho S, Yun Y, Jang S, Ra Y, Choi JH, Hwang HJ, Choi D, Choi D (2020) Universal biomechanical energy harvesting from joint movements using a direction-switchable triboelectric nanogenerator. Nano Energy 71:104584
28. Fan F-R, Tian Z-Q, Wang ZL (2012) Flexible triboelectric generator. Nano energy 1(2):328–334
29. Li S, Wang J, Peng W, Lin L, Zi Y, Wang S, Zhang G, Wang ZL (2017) Sustainable energy source for wearable electronics based on multilayer elastomeric triboelectric nanogenerators. Advanced Energy. Materials 7(13):1602832
30. Bairagi S, Ali SW (2020) Investigating the role of carbon nanotubes (CNTs) in the piezoelectric performance of a PVDF/KNN-based electrospun nanogenerator. Soft Matter 16(20):4876–4886
31. Xu Z, Jin C, Cabe A, Escobedo D, Hao N, Trase I, Closson AB, Dong L, Nie Y, Elliott J (2020) Flexible energy harvester on a pacemaker lead using multibeam piezoelectric composite thin films. ACS Appl Mater Interfaces 12(30):34170–34179
32. Shi K, Huang X, Sun B, Wu Z, He J, Jiang P (2019) Cellulose/BaTiO3 aerogel paper based flexible piezoelectric nanogenerators and the electric coupling with triboelectricity. Nano Energy 57:450–458
33. Zhou H, Zhang Y, Qiu Y, Wu H, Qin W, Liao Y, Yu Q, Cheng H (2020) Stretchable piezoelectric energy harvesters and self-powered sensors for wearable and implantable devices. Biosens Bioelectron 168:112569
34. Liu H, Lin X, Zhang S, Huan Y, Huang S, Cheng X (2020) Enhanced performance of piezoelectric composite nanogenerator based on gradient porous PZT ceramic structure for energy harvesting. J Mater Chem A 8(37):19631–19640
35. Bafghi ZG, Manavizadeh N (2020) Low power ZnO nanorod-based ultraviolet photodetector: Effect of alcoholic growth precursor. Opt Laser Technol 129:106310
36. Tsai SY, Chen CC, Huang J-M, Lai Y-S, Ku C-S, Lin C-M, Ko F-H (2021) Piezo-enhanced Thermoelectric Properties of Highly Preferred c-Axis ZnO Nanocrystal Films: Implications for Energy Harvesting. ACS Appl Nano Mater 4(9):9430–9439
37. Chowdhury AR, Abdullah AM, Hussain I, Lopez J, Cantu D, Gupta SK, Mao Y, Danti S, Uddin MJ (2019) Lithium doped zinc oxide based flexible piezoelectric-triboelectric hybrid nanogenerator. Nano Energy 61:327–336
38. Cao X, Xiong Y, Sun J, Zhu X, Sun Q, Wang ZL (2021) Piezoelectric Nanogenerators Derived Self-Powered Sensors for Multifunctional Applications and Artificial Intelligence. Adv Funct Mater 31(33):2102983
39. Yang Y, Pan H, Xie G, Jiang Y, Chen C, Su Y, Wang Y, Tai H (2020) Flexible piezoelectric pressure sensor based on polydopamine-modified BaTiO3/PVDF composite film for human motion monitoring. Sens Actuators A: Phys 301:111789
40. Rezaie S, Bafghi ZG, Manavizadeh N, Kordmahale SB (2021) Highly Sensitive Detection of Dissolved Gases in Transformer Oil With Carbon-Doped ZnO Nanotube: A DFT Study. IEEE Sens J 22(1):82–89
41. Le AT, Ahmadipour M, Pung S-Y (2020) A review on ZnO-based piezoelectric nanogenerators: Synthesis, characterization techniques, performance enhancement and applications. J Alloy Compd 844:156172
42. Ma J, Ren J, Jia Y, Wu Z, Chen L, Haugen NO, Huang H, Liu Y (2019) High efficiency bi-harvesting light/vibration energy using piezoelectric zinc oxide nanorods for dye decomposition. Nano Energy 62:376–383
43. Farajollahi H, Bafghi ZG, Mohammadi E, Manavizadeh N, Salehi A (2020) Sensitivity enhancement of AZO-based ethanol sensor decorated by Au nano-islands. Curr Appl Phys 20(8):917–924
44. Rao J, Chen Z, Zhao D, Yin Y, Wang X, Yi F (2019) Recent progress in self-powered skin sensors. Sensors 19(12):2763
45. Afshari F, Golshan Bafghi Z, Manavizadeh N (2022) Unsophisticated one-step synthesis super hydrophilic self-cleaning coating based on ZnO nanosheets. Appl Phys A 128(1):1–9

46. Hamid HMA, Çelik-Butler Z (2018) Characterization and performance analysis of Li-doped ZnO nanowire as a nano-sensor and nano-energy harvesting element. Nano Energy 50:159–168
47. Hao N, Xu Z, Nie Y, Jin C, Closson AB, Zhang M, Zhang JXJ (2019) Microfluidics-enabled rational design of ZnO micro-/nanoparticles with enhanced photocatalysis, cytotoxicity, and piezoelectric properties. Chem Eng J 378:122222
48. Zhang X, Le M-Q, Zahhaf O, Capsal J-F, Cottinet P-J, Petit L (2020) Enhancing dielectric and piezoelectric properties of micro-ZnO/PDMS composite-based dielectrophoresis. Mater Des 192:108783
49. Qian Y, Kang DJ (2018) Poly (dimethylsiloxane)/ZnO nanoflakes/three-dimensional graphene heterostructures for high-performance flexible energy harvesters with simultaneous piezoelectric and triboelectric generation. ACS Appl Mater interfaces 10(38):32281–32288
50. Shakthivel D, Dahiya AS, Mukherjee R, Dahiya R (2021) Inorganic semiconducting nanowires for green energy solutions. Curr Opin Chem Eng 34:100753
51. Wang J, Qian S, Yu J, Zhang Q, Yuan Z, Sang S, Zhou X, Sun L (2019) Flexible and wearable PDMS-based triboelectric nanogenerator for self-powered tactile sensing. Nanomaterials 9(9):1304
52. Parida K, Thangavel G, Cai G, Zhou X, Park S, Xiong J, Lee PS (2019) Extremely stretchable and self-healing conductor based on thermoplastic elastomer for all-three-dimensional printed triboelectric nanogenerator. Nat Commun 10(1):1–9
53. Dong K, Peng X, Wang ZL (2020) Fiber/fabric-based piezoelectric and triboelectric nanogenerators for flexible/stretchable and wearable electronics and artificial intelligence. Adv Mater 32(5):1902549
54. Kannan JA, Balasubramanian K (2020) Thermally influenced, optical and fluorescence properties of Zinc Oxide nanoparticles for glutathione sensing. Appl Phys A 126(8):1–11
55. Markova-Velichkova M, Veleva S, Tumbalev V, Stoyanov L, Nihtianova D, Mladenov M, Raicheff R, Kovacheva D (2013) XRD and TEM characterization of the morphology of ZnO powders prepared by different methods. Bulg Chem Commun 45(4):427–433
56. Wang X, Yang B, Liu J, Yang C (2017) A transparent and biocompatible single-friction-surface triboelectric and piezoelectric generator and body movement sensor. J Mater Chem A 5(3):1176–1183
57. Nourafkan M, Mohammadi E, Manavizadeh N (2019) Influence of the ZnO nanostructures shape on piezoelectric energy harvesters performance. IEEE Trans Electron Devices 66(11):4989–4996
58. Yang T, Pan H, Tian G, Zhang B, Xiong D, Gao Y, Yan C, Chu X, Chen N, Zhong S (2020) Hierarchically structured PVDF/ZnO core-shell nanofibers for self-powered physiological monitoring electronics. Nano Energy 72:104706

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.